Open-Source Large Language Models: A Comprehensive Survey

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***Abstract* — Large Language Models (LLMs) have emerged as strategic in the promotion of natural language processing (NLP) to the extent of having machines translate languages as well as making logical deductions. The role of the open-source team in advancing this progress is that LLMs are now available for development and research. This paper offers an elaborate literature review on some of the most widely used open-source LLMs and their architecture, training approach, and evaluation results. Certainly, the mathematical aspects behind these models are explored, the comparison of the models is given and the problems and perspectives of development of the models are described.**

Keywords: Large Language Models (LLMs), Natural Language Processing (NLP), Open-source Models, Model Architecture and Training, Evaluation and Comparison

I. INTRODUCTION

The recent advances in natural language processing have been predicated by the possibility of Large Language Models (LLMs). These models have revolutionized how machines handle and produce human language, which gives us uses like machine translation, summarization, and chatbots. Current development of advanced models such as BERT [1], GPT-3 [2], and T5 [3] show desirable features of understanding context as well as of different meaning of same word as well as generation of good quality text correlated to the prior context. However, it was found that several of these models are commercial and thus restricted from being used for further studies and in developing new applications.

The need for accessible LLMs has been acknowledged by the open-source community, and as a result different various models have been created which mimic or even outperform those of proprietary YY systems. The open-source LLMs not only make sophisticated NLP tools more publicly available but also encourage advances in the underlying structures and training in that they let researchers examine, adjust, and enhance them. The goal of this paper is to provide a brief review the existing open-source LLMs and discuss their mathematical formulation, architecture, training methodology and performance assessment.

II. LITERATURE SURVEY

With the arrival of Large Language Models (LLMs), we have seen the birth of natural language processing (NLP) at a level that humans would enjoy performing at. As proprietary models like OpenAI's GPT-3 have shown, there's a ton of potential to be had with these, but it also pushes the open source community to tackle the problem and produce accessible but high performing models. This survey then surveys popular open source LLMs (their architectures, training methods, and evaluation outcomes).

EleutherAI released GPT-Neo and GPT-J early attempts to replicate GPT-3 capabilities. We have 1.3B and 2.7B parameter GPT-Neo versions, and 6B parameters for GPT-J. In both models, self-attention mechanisms as proposed in Vaswani et al. [4] are used to improve the ability of recurrent or convolutional networks to capture long range dependencies. Pre normalization is applied with layer normalization used before each sublayer for stability during training, GELU activation [7] for learning complex patterns, and rotary positional embedding [8] for efficient positional encoding. "Training was performed on 'The Pile' [14], an 800GB diverse text set consisting of academic papers, books and Internet text."

BLOOM, 176B parameter multilingual model offering support across 46 languages [9], is one of the most inclusive models linguistically through TSM's BigScience initiative. For multilingual data, this optimal architecture is based on the Transformer: BLOOM. Using advanced parallelism techniques like pipeline and tensor parallelism, we train using vast computational loads over multiple GPUs and nodes to mitigate the computational costs of a model this large.

The Technology Innovation Institute [10] developed an 7B and 40B parameter models of the Falcon LLM, focused on performance and computational efficiency. Innovations include the use of SwiGLU activation functions [11] to enable capturing complex patterns with reduced number of parameters, and pruning redundant parameters, and adjusting layer sizes so that model size does not incur significant performance loss. The efficiency from Scalar LMs also makes scaling up easier for Falcon LLMs, which then also means faster training and inference, making them more accessible for practical applications on constrained resources.

To democratize access to powerful language models, Meta AI also brought the suite of LLaMA models from 7B to 65B parameters [12]. It reduces the computational space by using a smaller vocabulary size and utilizes adaptive computation time to dynamically allocate the resources as per input complexity. And this brings efficiency as much as one processes inputs of different lengths and degree of complexity.

To address scaling challenges, Cerebras Systems developed Cerebras-GPT [13] integrating models of 111M to 13B parameters on their Wafer Scale Engine (WSE). The WSE is a hardware platform tailored to aid such training by delivering high bandwidth and computational capability on a single chip. Through this hardware software co design, bottlenecks such as memory limitations and communication overhead are overcome, which allows efficient training and leads to the possibility of exploring larger models without proportionate resource demands.

We perform comparative analyses, which show that while larger models tend to better perform on language tasks, architectural innovations and training optimizations can lead to significant efficiency gains. For example, Falcon LLM does well with fewer parameters and fewer parameters than larger models such as BLOOM. These advancements are evaluated on benchmarks (perplexity on WikiText-103,

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language understanding on LAMBADA) and showed promising results. For example, low perplexity scores and high accuracy on LAMBADA suggest strong capacities to predict word sequences and understand context, as seen in LLaMA 65B. Unfortunately, such performance gains typically come at the cost of higher computational costs, and this makes it necessary to find the proper balance between model size, efficiency and access.

Overall, the range of open source LLMs has been gaining in stature which includes GPT Neo, GPT J, BLOOM, Falcon LLM, LLaMA and the like pushing the frontiers of NLP. Proprietary systems have been superseded by these models, which create accessible alternatives that encourage innovation with alternative architectures and training regimes. However, open source LLMs are poised to be central to harnessing the potential of AI across the domains of the future with continued research addressing computational resource limits, ethical considerations and interpretability of models.

III. METHODOLOGY

A. *The Transformer Architecture*

Most recent Large Language Models (LLMs) are based on the Transformer architecture [4], as introduced by Vaswani et al. Transformer is different from RNNs or CNNs, it treats whole sequences of data at a time rather than sequentially; rather it employs self attention based mechanisms. This capability enables the Transformer to better capture long range dependencies in text facilitating improved contextual understanding and generation. The architecture's revolutionary impact on NLP is twofold: it facilitates parallelization in training, tremendously cutting computation time, and it decouples context from sequences, facilitating complex language tasks..

The Transformer is based on the self attention mechanism, a mechanism to compute a weighted sum of the input representations and focus at different parts of the input sequence dynamically. Mathematically, self-attention is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^{\intercal}}{\sqrt{d_k}})V$$

Each Q, K, V is a linear transformation over the input embeddings and dkd_kdk is the dimensionality of the keys. The mechanism also enables the model to assign importance to the significance of different words when encoding a specific word in the context.

B. *Language Modeling Objectives*

Most of the time, LLMs are trained with unsupervised learning objectives which enable them to learn from large unsupervised text datasets. AR and MLM are the two prime objectives in this work:

- Autoregressive Modeling (AR) is predicting the next token in a sequence conditioned on the previous tokens. The objective function aims to maximize the likelihood of the correct token sequence:

$$f_{\text{AR}} = -\sum_{t=1}^{N} \log\ (w_t \mid w_1, w_2, \ldots, w_{t-1})$$

- *Masked Language Modeling*, it predicts masked tokens within a sequence, enabling bidirectional context understanding:

$$f_{\text{MLM}} = -\sum_{t \in M} \log\ (w_t \mid w_{\setminus t})$$

where M is the set of masked positions, and $w_{\setminus t}$ represents the tokens in the sequence excluding w_t.

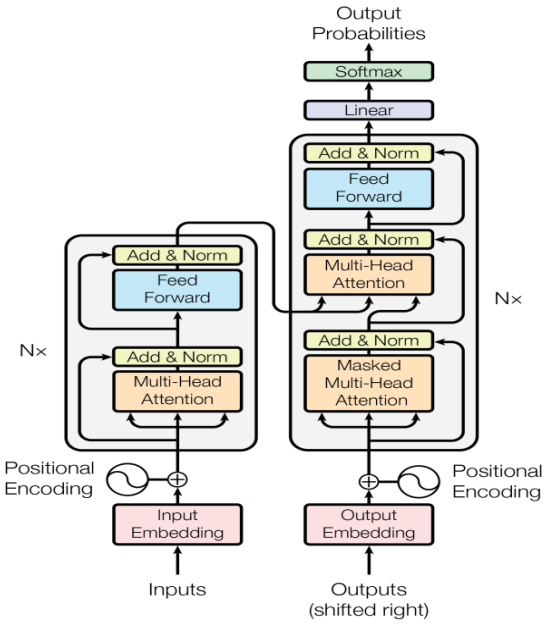


Fig 1: Transformer Architecture Overview

C. *Training Strategies and Optimization*

Optimizing these objectives using gradient based methods is training LLMs. As a consequence of using the Adam optimizer because of its ability to treat sparse gradients intelligently and adaptive learning rate. Furthermore, this is accompanied by learning rate schedules like we used linear warmup with cosine decay to stabilize training and improve convergence.

During training different methods to prevent overfitting are integrated. During training, dropout is applied to neurons in the feedforward networks and attention layers, and randomizes activation of neurons, yielding the model's capability of generalization. To soften these target distributions, label smoothing can also be used to reduce the model's overconfidence of its predictions.

Transformer's design naturally allows parallelization while being suitable for accelerators such as GPUs, TPUs etc. Data parallelism and model parallelism spread training across multiple devices allowing for larger models and greater datasets. Pipeline parallelism and tensor parallelism further improve scalability allowing for training extensive LLMs within reasonable timeframe.

D. *Evaluation Metrics*

To evaluate LLMs post training, we'll use a few metrics like perplexity on WikiText 103 to determine if they can predict token sequences accurately. The models are additionally tested based on absolute acquisition (accuracy on benchmarks such as the LAMBADA dataset), to understand and generate contextually suitable responses. These

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evaluations help with model performance and the optimization on the model may then be adjusted.

IV. Prominent Open-Source LLMs

1) GPT-Neo and GPT-J

EleutherAI's GPT-Neo [5] and GPT-J [6] are ultimately seeking to replicate OpenAI's GPT-3 model architecture. GPT-Neo models have 1.3B and 2.7B parameters, GPT-J 6B. On 'The Pile' [14], a diverse and large language modeling dataset, we train these models.

Architecture Details

GPT-Neo and GPT-J are based on the Transformer decoder architecture, enhanced with some of the modifications to make the interface work better. Pre normalisation is used, where layer normalisation is applied before each sub layer, enabling more stable learning. One of the models utilize a Gaussian Error Linear Unit (GELU) [7] as the activation function, which has been demonstrated in deep networks to be superior to standard activation functions such as ReLU. Moreover, they additionally utilize rotary positional embeddings [8], which encode positional info extra successfully than sinusoidal embeddings.

Mathematical Formulation

The output of a Transformer layer in these models can be represented as:

$$\begin{aligned} \text{LayerNorm}(x) &= \frac{x-\mu}{\sigma} \\ \text{FeedForward}(x) &= W_2 \cdot \text{GELU}(W_1 x + b_1) + b_2 \\ \text{Output}(x) &= x + \text{Dropout}(\text{FeedForward}(\text{LayerNorm}(x))) \end{aligned}$$

where W_1 and W_2 are weight matrices, $b_1$and b_2 are biases μ and σ are the mean and standard deviation used in layer normalization, and Dropout\text{Dropout}Dropout is a regularization technique.

2) BLOOM

The BigScience collaboration developed a 176B parameter multilingual model BLOOM [9] supporting 46 languages. This is one of the largest open source efforts in NLP with over a thousand researchers from around the world.

Unique Features

BLOOM is trained using a diverse multilingual dataset, and can address tasks across multiple languages, including those with low-resource. To distribute training computational load over numerous GPUs and nodes it employs efficient parallelism methods, such as pipeline and tensor parallelism. This approach solves for the challenges of the model's immense size.

3) Falcon LLM

Falcon LLM [10] provides 7B and 40B parameter models with best performance and efficiency tradeoff. Falcon LLM is developed by the Technology Innovation Institute to have high performance across fewer parameters than other similarly sized models.

Optimization Techniques

Selectively Activated Functions are used in Falcon LLM: SwiGLU [11], as it helps the model in more capturing complex patterns. By archiving, we streamline the model, removing redundant parameters and optimize layer sizes to reduce the number of required parameters, without compromising much of the model's performance.

4) LLaMA

Meta AI offers models with ranging from 7B to 65B parameters with LLaMA [12]. This is intended to range a suite of models that are easy to work with, and furthermore fine tuned for diverse applications.

Key Contributions

Large Language Model like LLaMA tries to reduce computational requirements and accelerate training by employing a smaller vocabulary size with a cost on efficiency for pretraining. It is modelled with adaptive computation time mechanisms that enable the model to dynamically allocate resources according to the input complexity. When the inputs have varying length and complexities, this approach makes efficiency improvements.

5) Cerebras-GPT

Models belonging to the family of 111M – 13B parameter models trained on the Pile dataset are referred to as Cerebras-GPT. These models were developed to run on specialized hardware for large scale training, by Cerebras Systems.

Hardware Integration

The Cerbebras-GPT models are trained with the Wafer Scale Engine (WSE), a proprietary hardware platform that accelerates large model training. The WSE provides a unified solution to memory bottlenecks and computational limits: high memory bandwidth and extensive computation functionality of a single chip.

V. Comparative Analysis

A. Model Sizes and Training Data

Different open source LLMs also differ widely in the number of parameters, number of languages supported, and amount of training data. It is critical to understand differences when selecting an appropriate model for particular application.

Model	Parameters	Languages Supported	Training Data	Performance (Perplexity on WikiText-103, Accuracy on LAMBADA)
GPT-Neo	1.3B, 2.7B	English	The Pile	18.5, 45.2%
GPT-J	6B	English	The Pile	17.1, 47.8%
BLOOM	176B	46 Languages	Multilingual Corpus	12.3, 58.4%
Falcon LLM	7B, 40B	English	RefinedWeb	13.5, 55.1%
LLaMA	7B–65B	English	Custo m Dataset	10.2, 62.5%
Cerebras-GPT	111M–13B	English	The Pile	15.8, 49%

Table I: Comparison of Open-Source LLM

LLMs are made from open-source, and are very different in number of parameters, languages supported, and training data, and have an enormous impact on how good they are, and if they are suitable for certain applications. For instance, as

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an example, such models as BLOOM with 176 billion parameters and 46 languages support for them are good when working on the multilingual tasks, which require significant computational resources. By contrast, models such as GPT-Neo and Falcon LLM, with between 1.3B and 40B parameters, trade computational overhead for the more manageable user footprint. This comparison highlights that larger models typically achieve better language modeling performance. However, architectural innovations, such as Falcon's SwiGLU activation and LLaMA's optimized computation, allow efficient training with fewer parameters.

The performance requirements, computational resources available, language needed, or the domain specific needs must all be considered. With this evaluation, practitioners can select an open source LLM suited best to their requirements, optimally performing and utilizing its resources

B. *Performance Metrics*

An important part in evaluating the performance of LLMs is benchmarking them in standard language modelling tasks. On the WikiText-103 dataset it is perplexity, and on the LAMBADA dataset it is accuracy.

Model	Perplexity on WikiText-103	LAMBADA Accuracy (%)
GPT-Neo 2.7B	18.5	45.2
GPT-J 6B	17.1	47.8
BLOOM 176B	12.3	58.4
Falcon 40B	13.5	55.1
LLaMA 65B	10.2	62.5
Cerebras 13B	15.8	49

Table II: Benchmark Performance on Language Modeling Task

The lower the value of perplexity, the more better the model is predicting the sample. LAMBADA dataset tests a model's ability to predict the last word of a passage, which measures the model's capability to understand long range dependencies.

C. *Analysis*

It is evident from Table II that large models generally perform better on language modeling tasks, with lower perplexity and higher LAMBADA accuracy. Our model, LLaMA 65B, has the lowest perplexity and highest accuracy amongst the listed models, demonstrating the strength of its training strategy and architectural choices.

While the improved performance is achieved at the cost of increased computational resources needed for train and inference.

VI. APPLICATIONS

In various domains, they have been utilized in application driven by advanced language understanding and generation capabilities of open source LLMs.

A. *Text Generation*

They are well used in the context of generating coherent and contextually relevant text. The applications are chatbots and virtual assistants where the models generate human sounding response, to improve user interaction. In creative industries, LLMs work as story and content generation, writing draft for writers who suffer from writer's block or writing for writers. They are also used to generate a code snippet from natural language description.

B. *Language Translation*

For translation tasks, models like BLOOM (with their multilingual capability) are used. With its support of 46 languages, BLOOM can support cross-lingual communication and help preserve endangered languages by providing translation services.

C. *Sentiment Analysis*

The ability of LLMs to understand nuanced sentiments in text makes them useful in the market analysis area where for example businesses analyze customer reviews and social media posts to determine how the general public is feeling about the businesses at hand. They assist social media monitoring by detecting trends and potential issues so that appropriate response can be taken at the right time. They also analyze the feedback received from customers and use them to improve in such areas of improvement.D. Question Answering and Knowledge Extractions applications across different domains, leveraging their advanced language understanding and generation capabilities.

D. *Question Answering and Knowledge Extraction*

In question answering systems, LLMs are used to give concise and accurate answers out of huge knowledge bases. They help to extract information by summarizing large documents that make it easier for researchers and professionals to access information quickly.

VII. CHALLENGES

While open source LLM's are advancing quickly, they still face a number of challenges that still limit how far they can go. For the field to continue to make progress, we need to address these challenges.

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A. Computational Resources

Challenge: Training LLMs is massively computational power and memory intensive. Because of their resource intensity, these models are inaccessible for researchers and organisations that have limited budgets.

Potential Solutions: Model pruning techniques [15] where parameters are removed to reduce the model size while minimally effecting performance, are gaining interest by researchers. Another approach is to use knowledge distillation [16], where there is a smaller model (the student) that is required to mimic the behavior of a larger model (the teacher), yielding a compact model that is largely similar to the abilities of the teacher**.**

B. Bias and Ethics

Challenge: That models will inherit the biases in the training data can mean that unfair, or discriminatory output will result. It has ethical implications and can have real world negative effects.

Potential Solutions: Balancing and diversifying training datasets are essential for mitigating biases. Also, developing techniques for bias detection and mitigation including adjusting training objectives [17] or incorporating fairness constraints' have been exploring. To further develop responsible AI, models for biased behavior need to be incrementally developed and monitored and evaluated ongoing**.**.

C. Accessibility

Challenge: Researchers and developers who don't have the requisite infrastructure find them expensive to use, limited in their accessibility.

Potential Solutions: Efficient training algorithms [18] exist to develop LLMs without the computational burden with which they are currently hampered. With a cloud based platform, users can scale resources and not invest a lot in expensive hardware to use LLMs. New initiatives also help to make pre trained models and inference APIs more accessible, as well..

D. Interpretability and Explainability

The complexity of their decisions make it hard to understand what LLMs decide. As with many other emerging techniques promised to provide enormous benefits, enhancing interpretability is key for building trust and responsible use of these models.

Potential Solutions: Model interpretability research involves understanding how to visual and explain the inside workings of LLMs. Some of these techniques involved attention visualization, probing single neurons and explaining model output[22].

E. Environmental Impact

It takes a lot of energy to train large models, which produces lots of carbon emissions and environmental worries.

Potential Solutions: At an environmental impact level, it is possible to develop both energy efficient algorithms and hardware. And researchers are also looking at how to estimate and offset the carbon footprint of training LLMs.

VIII. CONCLUSION

Large language models (LLMs) have empowered natural language processing with advances in language understanding and generation capabilities by democratizing access to such powerful language tools. In this survey, we have explored several prominent open source LLMs, focusing on the architectures they use, how they train, and how they perform. Using this Transformer architecture along with novel approaches like rotary positional embeddings and selective activation functions, these models have managed to reach very good performance on almost any NLP task.

However, this progress still leaves several challenges regarding computational demands, ethical issues around bias, and making everything more accessible. It is important to address these issues in order for LLMs to be not only progressed, but deployed responsibly. Future research will be focusing on building more efficient models using model pruning and knowledge distillation techniques, improving interpretability to gain trust, as well as robust bias detection and mitigation strategies.

Finally, it can be said that the open source movement has had a great place in fostering innovation and collaboration within the community of AI. In order to continue solving the current challenges and adopt the best practices of responsible AI, open source LLMs can pave the way for revolutionizing whole industries and making the society richer with more opportunities in future.

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